

AIVANA COMPREHENSIVE CANCER CARE OS & HOSPITAL MANAGEMENT SYSTEM

Master Functional Architecture, Module Specifications & Clinical Governance Guide

System Version: 1.0 Enterprise | **Status:** Verified & Passing (201/201 Automated Tests)

Clinical Norms: AJCC 8th Ed • NCCN/ESMO Guidelines • RECIST 1.1 • CTCAE v5.0 • DuBois BSA • CAP Synoptic Standards

Governance Standard: Human-in-the-Loop Confirmation • Mandatory Absence Vocabulary • Multi-Tenant RBAC

Executive Summary: Alvana OS is a unified clinical operating system and Hospital Management System (HMS) engineered for precision oncology centers and acute hospitals. It combines full hospital operations (Front Desk, Appointments, Billing, TPA Pre-Authorization, Pharmacy, Inventory, Nursing Wards) with an evidence-gated Cancer Care AI Operating System (CCA OS) that prevents autonomous diagnostic errors, enforces strict source provenance, and streamlines tumor board deliberations.

1. Precision Oncology Operating System (Alvana CCA OS)

The CCA OS is a 22-screen specialized oncology workflow deck built specifically for cancer treatment centers. It operates on strict clinical governance principles where AI assists but never acts autonomously.

- **96px Persistent Patient Header:** A continuous 2-row clinical identity bar across all screens displaying verified demographics, DuBois BSA (1.65 m²), ECOG Performance Status, and 6 active-glow status engine pills (Summary, Journey, Staging, NCCN Context, NEXUS Brief, Care Plan, Contradiction Alert).
- **Longitudinal Evidence Graph & 2-Click Provenance (DSC-03):** Every clinical fact (IHC receptors, tumor grade, margin status) is stored as an immutable atomic entity linked to verbatim source text, PDF page, and bounding box. The 2-click slide-over drawer lets oncologists instantly audit the original biopsy or scan.
- **Cross-Document Contradiction Detector (CTR-01):** Automated engine that flags discordance across multimodal records (e.g. referral says Left breast, biopsy says Right breast). Renders an amber alert blocking unreviewed treatment signing.
- **AJCC 8th Edition Staging State Machine:** Evidence-gated state transitions: EVIDENCE_INCOMPLETE → PARTIALLY_READY → READY_FOR_STAGING → CLINICIAN_CONFIRMED. Missing distant metastasis scans (CECT/PET) explicitly block autonomous staging until the clinician formally confirms.
- **Guideline Readiness Gate (Rule G-5):** Strict safety lock that hides NCCN/ESMO guideline recommendations until the treating oncologist has formally signed off on the verified AJCC stage, preventing premature anchoring bias.
- **NEXUS 13-Section Clinical Brief Synthesizer:** Deterministic multi-tier engine distilling 100-page historical cancer files into 13 structured sections: Identity, Diagnosis, Biomarkers, Staging, Treatment History, Comorbidities, Contradictions, Best Next Test, NCCN Alignment, MDT Topics, Qualitative Uncertainty, Toxicity Baselines, and Quality Measures.
- **1-Click Multi-Disciplinary Tumor Board (MDT) Package:** Bundles staging, imaging links, pathology scans, and clinical brief into a presentation-ready case package with multi-specialist consensus decision logging.
- **Multi-Modality Live Care Plan Engine (Rule E-36):** Pre-fills evidence-based regimens (e.g. Dose-Dense AC-T, Lumpectomy+SLNB, RT, Endocrine). Enforces immutable versioning where any medication or cycle modification requires mandatory change rationale logging.

- **Daycare Chemotherapy Clearance Cockpit:** Treatment-day cockpit evaluating real-time CBC/Diff (ANC ≥ 1500 , Platelets $\geq 100k$), renal/liver panels, and CTCAE toxicities with 5 structured clearance exits: Standard, Dose Reduction, Regimen Switch, Hold, and Discontinue.
- **CTCAE v5.0 Toxicity Logger with Baseline Guard:** Standardized adverse event logging (Grades 1–5 for Neuropathy, Nausea, Mucositis, Diarrhea, Neutropenia) with automated alerts if current toxicity exceeds pre-treatment baselines.
- **RECIST 1.1 Longitudinal Response Tracker:** Solid tumor tracking measuring Target Lesion Sum of Diameters against baseline and nadir, classifying Complete Response (CR), Partial Response (PR, $\geq 30\%$ decrease), Stable Disease (SD), or Progression (PD).
- **Clinical Absence Vocabulary Enforcement:** Absence of data is never masked as normal. Unrecorded or pending parameters render explicit tokens: NOT_RECORDED, UNKNOWN, PENDING_REVIEW, CONTRADICTED.

Cancer Care Standards Enforced in CCA OS:

- **AJCC Cancer Staging Manual (8th Ed):** Deterministic cTNM / pTNM / ypN evaluation.
- **NCCN / ESMO Clinical Practice Guidelines:** Evidence-based systemic therapy pathways.
- **DuBois & DuBois BSA Standard:** Strict mathematical formula ($0.007184 \times H^{0.725} \times W^{0.425}$).
- **NCI Common Terminology Criteria for Adverse Events (CTCAE v5.0):** Adverse event grading.
- **RECIST 1.1 Criteria:** Response Evaluation Criteria in Solid Tumors.

2. Outpatient Department (OPD) & Ambient AI Scribe

The OPD module accommodates rapid, structured clinical encounters, integrating ambient voice capture, fuzzy terminology validation, and real-time clinical decision support.

- **Structured SOAP Consultation Charting:** Comprehensive encounter charting capturing Chief Complaints, History of Present Illness (HPI), Objective Findings, Primary & Differential Diagnoses, Medications, Lab Orders, and Follow-up Advice.
- **Multilingual Ambient Audio Scribe:** Direct audio recording converting conversational clinician-patient dialogue into structured draft clinical notes with Groq/Sarvam transcription pipelines.
- **Clinical Decision Support (CDS) Allergy Conflict Engine:** Real-time safety engine that cross-references newly prescribed medications against the patient's recorded drug allergies (e.g. flagging Penicillin derivatives in allergic patients).
- **Fuzzy Formularies & Lab Test Name Matchers:** Deterministic string matchers that standardize free-text drug and laboratory order names against hospital master catalogs, catching clinical typos without generative hallucinations.
- **Longitudinal Follow-up Consultation Chaining:** Links follow-up consultations to previous index visits (follow_up_of_id) to reconstruct disease progression and therapy response trajectories.

Clinical Norms & Safety Standards in OPD:

- **SOAP Clinical Documentation Standard:** Structured Subjective, Objective, Assessment, Plan format.
- **Real-Time Allergy CDS Check:** Prevents anaphylaxis and severe adverse drug reactions.
- **Deterministic Formulary Normalization:** Prevents prescribing unapproved or ambiguous brand names.

3. Inpatient Department (IPD) & Advanced Nursing Charting

The IPD and Nursing module manages inpatient beds, DayCare oncology admissions, specialized risk scoring, fluid balance, and high-alert medication administration records.

- **Ward Capacity & Bed Assignment Management:** Real-time occupancy tracking, bed allocation, patient transfer workflows, and duplicate active admission conflict prevention.

- **DayCare Short-Stay Chemotherapy Admissions:** Fast-track admission workflow for same-day cancer chemotherapy infusion and blood transfusions without full inpatient bed allocation overhead.
- **Admission Baseline Nursing Assessments:** Standardized initial nursing intake documenting baseline vital signs, chief nursing complaints, physical examination, and initial risk stratification.
- **Morse Fall Risk Assessment:** 6-parameter fall scoring (History of falls, Secondary diagnosis, Ambulatory aids, IV therapy, Gait, Mental status) categorizing Low (0–24), Moderate (25–44), and High Risk (≥ 45) with automated precaution alerts.
- **Braden Scale for Pressure Ulcer Prediction:** 6-subscale evaluation (Sensory perception, Moisture, Activity, Mobility, Nutrition, Friction/Shear) stratifying Severe (≤ 9), High (10–12), Moderate (13–14), and Mild (15–18) pressure sore risk.
- **Numerical & Wong-Baker Pain Assessment:** Standardized 0–10 pain scoring with mandatory follow-up reassessment logging at 30 minutes (for IV analgesics) or 60 minutes (for Oral analgesics).
- **Intake/Output & 24-Hour Net Fluid Balance:** Tracks oral fluids, enteral feeds, and IV crystalloids/colloids against urine, surgical drains (JP/Hemovac/Chest tubes), and emesis, auto-calculating 24-hour cumulative net balance.
- **Electronic Medication Administration Record (MAR):** Enforces the 5 Rights of Medication Administration (Right Patient, Right Drug, Right Dose, Right Route, Right Time) with Given, Held, Refused, and Omitted status tracking.
- **Dual-Nurse Verification Signoff for High-Alert Drugs:** Mandatory second-nurse credential verification for cytotoxic chemotherapy, insulin, and concentrated electrolytes.
- **Bedside & Minor Procedure Documentation:** Charts bedside procedural notes, pre/post vitals, surgeon ID, and complications for paracentesis, thoracentesis, biopsies, and bone marrow aspirates.

Nursing & Inpatient Standards in IPD:

- **ISMP High-Alert Medication Double-Check:** Dual independent verification before chemo infusion.
- **Braden Scale for Pressure Injury Prevention:** Standardized skin integrity risk prediction.
- **Morse Fall Scale:** Evidence-based inpatient fall reduction protocol.
- **Hydration Safety Monitoring:** Strict hourly urine rate (target ≥ 0.5 mL/kg/hr) to protect kidneys.

4. Front Desk, Registration & Queue Management

Manages patient demographics, national identity numbers, appointment scheduling, and outpatient clinic flow.

- **Demographic Intake & ID Proof Capture:** Registers patients with mandatory demographic fields, national IDs (Aadhaar, PAN), contact details, and documented allergy history.
- **Doctor Appointment Scheduling & Conflict Prevention:** Books consultations with slot validation that prevents accidental double-booking of doctors.
- **Daily OPD Queue Tokens & ER Emergency Jump:** Generates daily sequential queue tokens with emergency priority override ('ER Workflow') to escalate critical oncology patients.
- **Patient Soft-Delete Lifecycle:** Administrative deletion endpoint (DELETE /api/admin/patients/{id}) that flags patient status as 'Deleted', maintaining historical data integrity while excluding deleted records from active search indices.

5. Revenue Cycle, Billing & TPA Pre-Authorization

Handles hospital tariffs, treatment packages, automated itemized billing, and insurance pre-authorization.

- **Dynamic Tariff Master & Treatment Packages:** Configurable charge master with custom packaged treatment discounts for surgical procedures and chemo cycles.
- **Consolidated Itemized Invoicing:** Auto-aggregates OPD consultations, bedside procedures, pharmacy dispensations, and ward bed charges into itemized invoices.

- **Multi-Mode Payment Settlement & Refund Ledger:** Collects cash, card, UPI, and insurance payments with partial payment tracking and authorized refund ledgers.
- **TPA Point-in-Time Clinical Snapshotting:** Generates a frozen JSON snapshot of patient demographics, diagnoses, allergies, consultations, and procedures at submission time, ensuring subsequent medical record updates cannot alter what was transmitted to the insurer.
- **Insurance Claim & Pre-Auth Approval Guard:** Enforces that insurance claims can only attach pre-authorizations that are in 'Approved' status, preventing claim rejections.

6. Pharmacy Dispensing & Inventory Supply Chain

Controls pharmaceutical dispensing, narcotics statutory registers, purchasing, and departmental stock movement.

- **Prescription Fulfillment & FIFO Batch Selection:** Inpatient and outpatient dispensing queue enforcing First-In-First-Out (FIFO) batch deduction and expiry checks.
- **Controlled Drugs (Narcotics / Schedule X) Register:** Statutory electronic register tracking dispensing of controlled analgesics (Morphine, Fentanyl) with real-time balance reconstruction.
- **Procurement & Purchase Order (PO) Pipeline:** Manages Vendor directories, Purchase Requests, Purchase Orders, and Goods Receipt Notes (GRN) with receipt stock incrementation.
- **Atomic Inter-Departmental Stock Transfers:** Race-condition-safe stock transfer mechanism moving inventory between Central Stores, Pharmacy, and Ward sub-stocks without risk of negative inventory.

7. Administration, Multi-Tenant RBAC & Audit Trails

Provides multi-tenant security isolation, role-based access control, and complete legal audit trails.

- **Multi-Tenant Isolation (organization_id):** Every database query is strictly scoped by the authenticated user's organization_id, preventing data leakage across clinics.
- **Role-Based Access Control (RBAC):** Fine-grained permissions for Admin, Doctor, HeadNurse, Nurse, NursingStation, Billing, TPA, Pharmacist, and InventoryManager.
- **Tamper-Proof Audit Logging (log_audit):** Systematically records timestamp, user ID, role, resource, action, and outcome for every clinical and administrative event.

8. Master Oncology Norms & Standards Compliance Matrix

The following matrix summarizes how every module in Alvana OS satisfies established oncology and healthcare norms:

Module	Clinical Norm / Standard	Governance & Enforcement Mechanism
CCA Staging	AJCC 8th Edition TNM	Evidence-gated state machine with mandatory doctor confirmation gate (staging.confirm)
CCA Guidelines	NCCN / ESMO Practice Guidelines	Rule G-5 locking pathway display until staging is clinician-confirmed
CCA Biometrics	DuBois & DuBois BSA Standard	Deterministic formula: $0.007184 \times H^{0.725} \times W^{0.425}$ (158cm/64kg = 1.65 m²)
CCA Provenance	CAP / Digital EMR Provenance	DSC-03: 2-click drill-down linking facts to verbatim text, page, and bounding box

Module	Clinical Norm / Standard	Governance & Enforcement Mechanism
CCA Toxicity	NCI CTCAE v5.0	Structured Grade 1–5 toxicity grading with baseline comparison safety alert
CCA Response	RECIST 1.1 Criteria	Automated Sum of Diameters comparison classifying CR, PR, SD, or PD
CCA Care Plan	ASCO / NCCN Chemotherapy Safety	Rule E-36 versioning with mandatory change-reason rationale logging
IPD Nursing MAR	ISMP 5 Rights & Double-Check	5 rights tracking + second nurse credential verification for chemo/high-alert drugs
IPD Risk Scores	Morse & Braden Standards	Standardized Fall Risk (0–125) and Pressure Ulcer (6–23) prediction tools
OPD Scribe & CDS	Allergy CDS & SOAP Standard	Real-time drug-allergy conflict detection + structured SOAP charting
Pharmacy	Narcotics / Schedule X Compliance	Controlled substance digital register with mathematical balance reconstruction
Billing / TPA	Insurance Pre-Auth Snapshotting	Frozen point-in-time clinical snapshotting + Approved status validation
Platform Auth	HIPAA / NABH Data Privacy	Multi-tenant organization_id scoping + PBKDF2 hashing + audit logging

9. Verification & Automated Test Certification

The entire platform has undergone rigorous automated unit, integration, concurrency, and workflow testing. All **201 test cases across 12 test suites** pass with a **100% success rate** in 11 minutes:

- **Appointments & Queue:** 18/18 Passed
 - **Billing & Invoicing:** 25/25 Passed
 - **TPA Pre-Authorization:** 44/44 Passed
 - **Inventory & Supply Chain:** 19/19 Passed
 - **Pharmacy & Controlled Drugs:** 24/24 Passed
 - **Nursing Assessments (Fall/Braden/Pain):** 14/14 Passed
 - **Nursing Charting (Fluid Balance / IV):** 10/10 Passed
 - **Medication Administration Record (MAR):** 8/8 Passed
 - **Procedures & Follow-up Linking:** 14/14 Passed
 - **SOAP Clinical Notes & Allergy CDS:** 11/11 Passed
 - **Deterministic CCA Engines:** 4/4 Passed
 - **9-Act CCA Oncology Live Workflow:** 10/10 Passed
- Total: 201 Passed, 0 Failed, 0 Regressions.**